Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details
Batch Number: 13

Start date:

Names: P Sahana sree

Internship Duration: 8 Weeks

1. Project Title

Stock TrendAI Stock Market Forecasting Model

2. Project Objective

The objective of this project is to develop an intelligent system capable of predicting stock market prices using historical financial data and deep learning techniques.
The project aims to analyze stock market trends and forecast future stock prices using a **Long Short-Term Memory (LSTM) neural network**, which is well-suited for time-series data analysis.

Another key objective is to design an **interactive visualization dashboard** using Streamlit that allows users to explore stock price movements, prediction results, and model performance metrics.
The dashboard provides a user-friendly interface for understanding stock behavior through graphical representations and statistical analysis.
This project demonstrates how **Artificial Intelligence and Machine Learning can be applied to financial data analysis** to assist in identifying patterns and predicting future market trends.

3. Project description in detail

The project implements a **stock price prediction model using LSTM neural networks**. Historical stock data is collected from **Yahoo Finance using the yfinance Python library**.

The dataset is preprocessed by handling missing values and scaling the data using **MinMaxScaler** to normalize the values between 0 and 1.

The data is then converted into **time-series sequences of 60 previous days**, which are used as input for the LSTM model to predict the next day's stock price.

After training the model, it is saved as **final_model.h5** and integrated into a **Streamlit-based dashboard**. The dashboard displays stock price trends, prediction graphs, and performance metrics such as **MSE, RMSE, and R^2 Score**.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Understanding the project requirements and researching stock market prediction techniques.	Studied time-series forecasting methods and explored machine learning approaches for financial data analysis.
Week 2	Collecting historical stock market data.	Used the Yahoo Finance API to download stock price data and explored the dataset structure.
Week 3	Data preprocessing and feature selection	Handled missing values, selected closing price for prediction, and normalized the dataset.
Week 4	Developing the deep learning model	Implemented the LSTM neural network architecture using TensorFlow and KerasDeve
Week 5	Training and optimizing the model	Trained the LSTM model using historical stock price sequences and evaluated model performance.
Week 6	Model evaluation and testing	Tested the model on unseen data and calculated performance metrics such as MSE and R^2 score.
Week 7	Dashboard development	Developed an interactive Streamlit dashboard for visualizing stock prices and predictions.
Week 8	Final testing and documentation	Completed final testing, prepared documentation, and finalized the project presentation

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Understanding project goals and technologies.	Week1
Prototype/First Draft	Initial model implementation.	Week 3
Mid-Term Review	Model training and evaluation.	Week 5
Final Submission	Completed dashboard and project report.	Week 7
Presentation	Final demonstration of the project.	Week 8

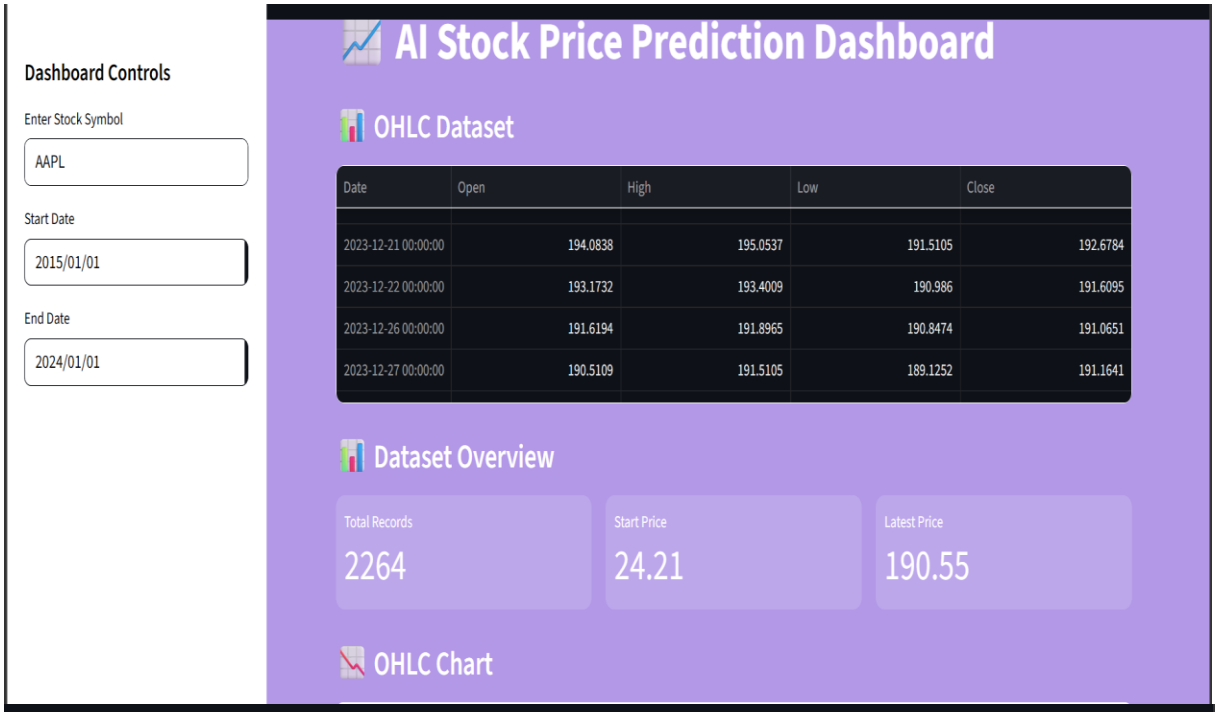
5b. Project execution details

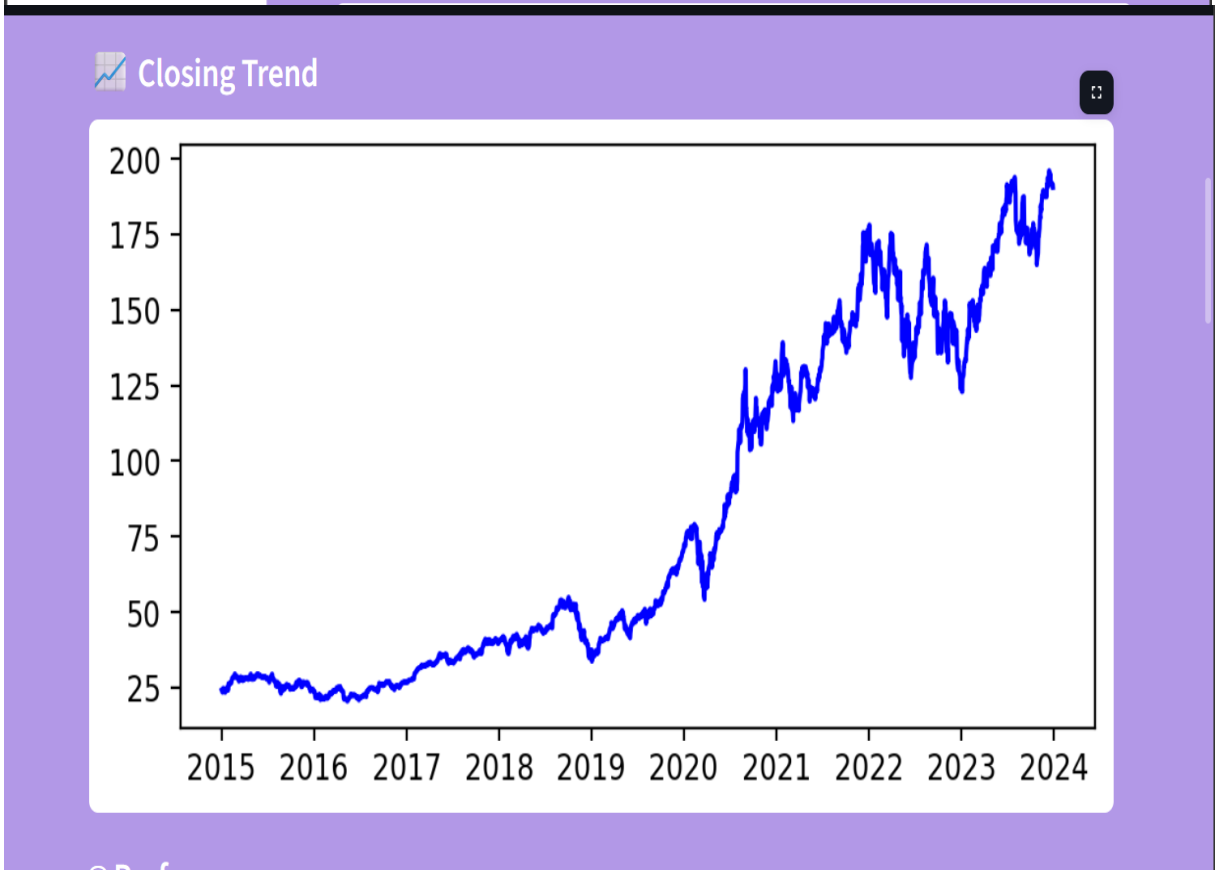
The project was executed in multiple stages. First, historical stock market data was collected using the **yfinance API**. The dataset included daily stock price information such as open, high, low, and closing prices.

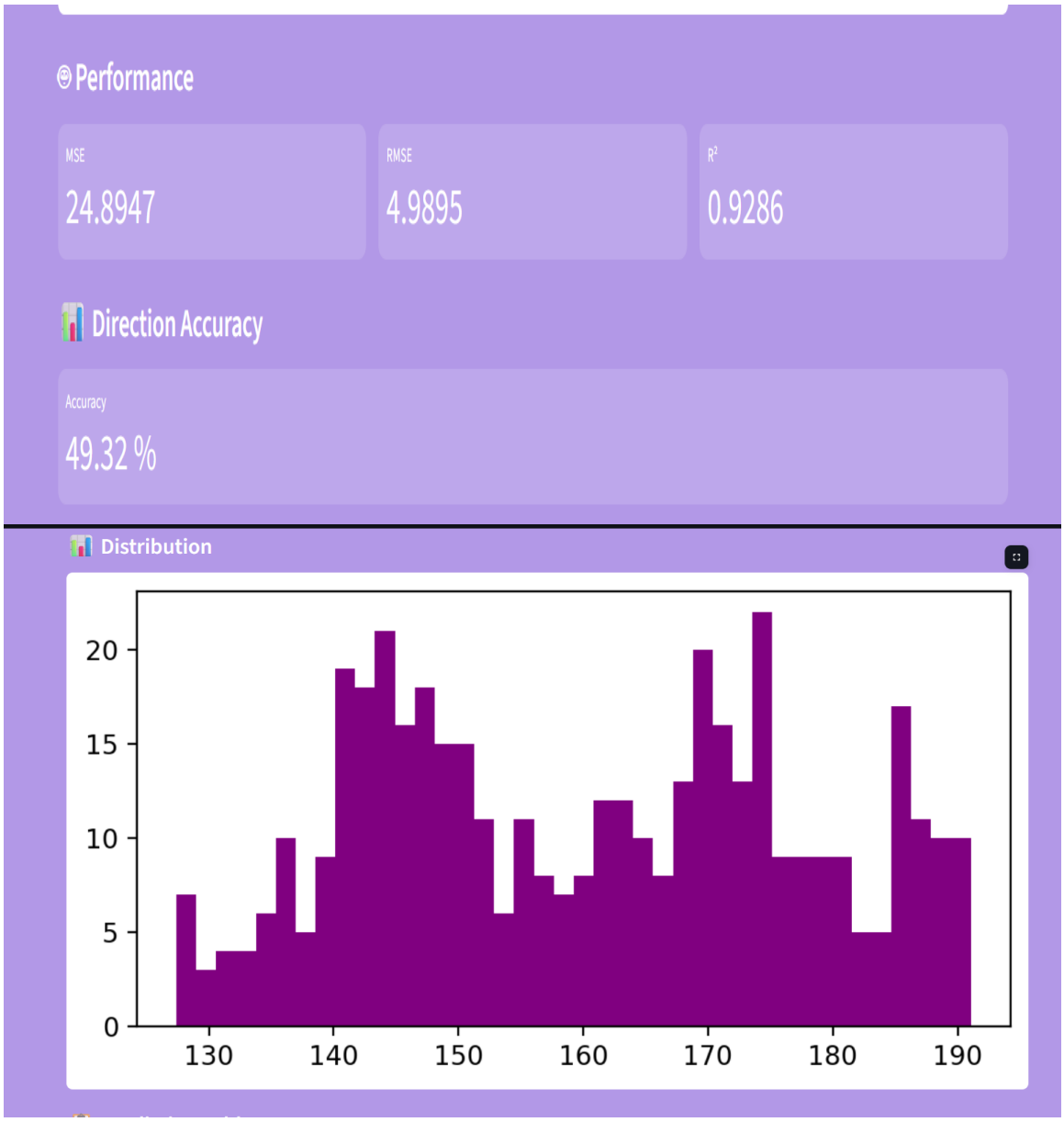
Next, the data was preprocessed by handling missing values and scaling the data using **MinMaxScaler**. The dataset was then converted into **time-series sequences**, where the model uses the previous 60 days of stock prices to predict the next day's price.

An **LSTM neural network model** was built using TensorFlow and Keras and trained using the processed dataset. After training, the model was saved and integrated into a **Streamlit dashboard**, which allows users to visualize stock price trends, prediction results, and model performance metrics.

6. Snapshots/Screenshots:



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RMSE
4.9895
R²
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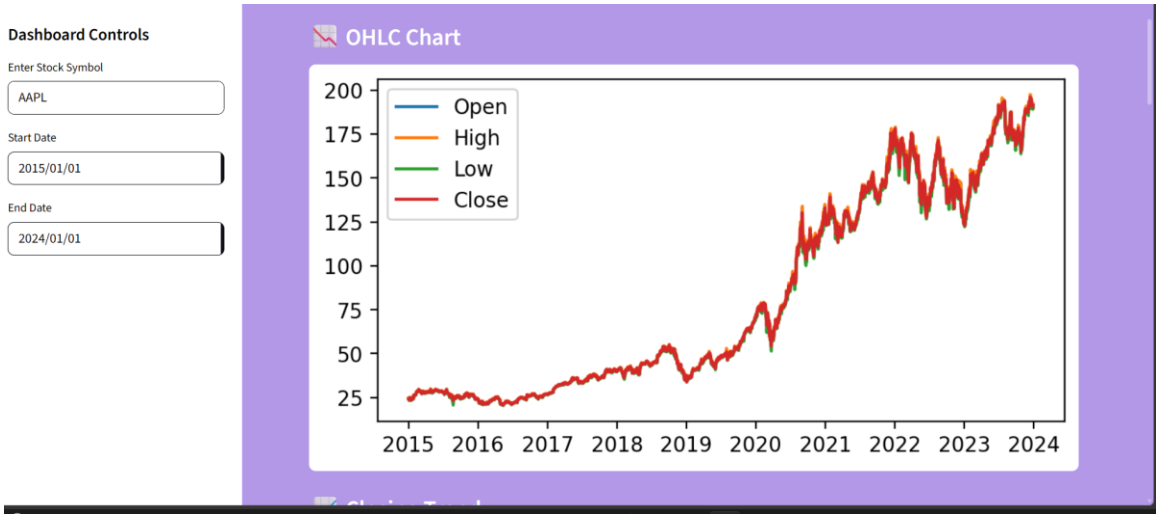
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OHLC Chart
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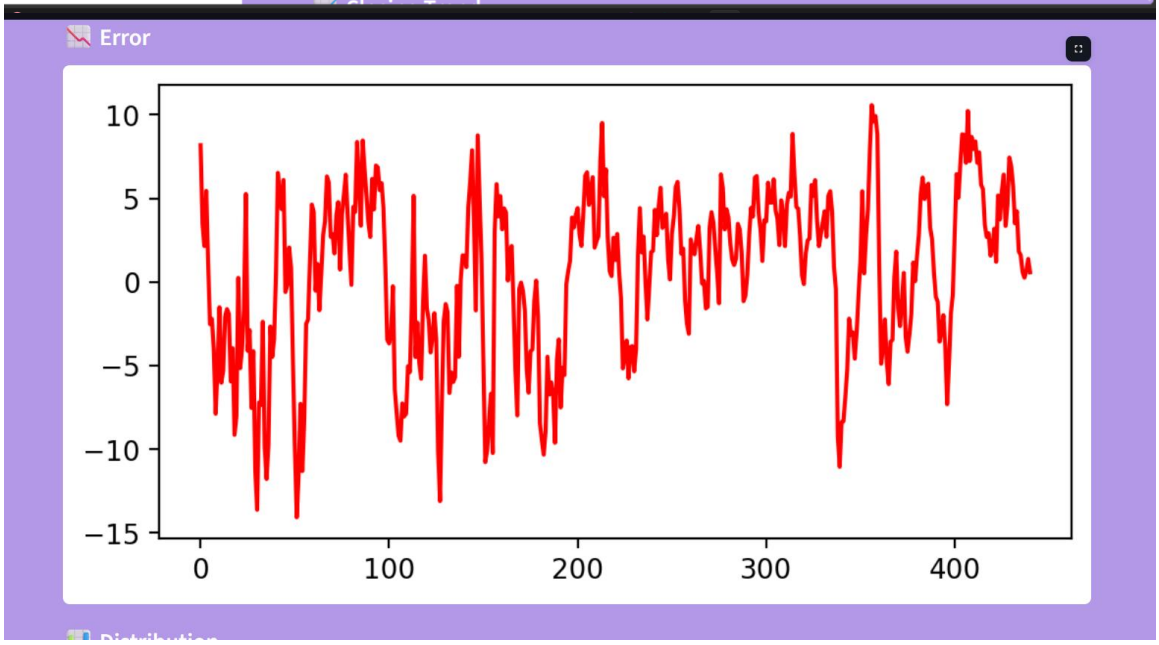
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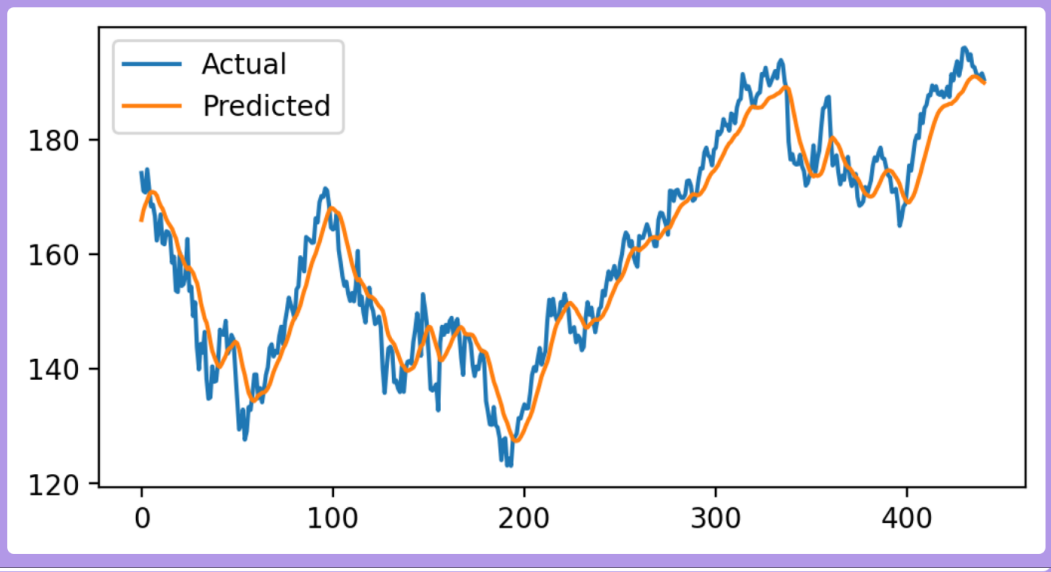
📈 Actual vs Predicted



📋 Prediction Table

	Actual	Predicted
0	174.1763	165.9964
1	171.0802	167.5958
2	170.7862	168.6343
3	174.8328	169.3833
4	171.5211	170.3663

📌 Insights

Trend: DOWNTREND 📉
R^2 Score: 0.929
Avg Error: 4.17

7. Challenges Faced

• Understanding the **structure of financial time-series data** and preparing it for machine learning models.

• Handling **missing values and inconsistencies in stock market data**.

• Converting raw stock data into **time-series sequences suitable for LSTM models**.

• Selecting appropriate **model parameters such as epochs, batch size, and sequence length**.

• Managing **model training time and computational resources**.

• Integrating the trained model with the **Streamlit dashboard for real-time visualization**.

• Debugging errors during **data preprocessing and model prediction stages**.

• Ensuring the dashboard displays **clear and meaningful visualizations for prediction analysis**.

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8. Learnings & Skills Acquired

- Python programming for data analysis and machine learning
- Data preprocessing using **Pandas and NumPy**
- Time-series forecasting techniques
- Deep learning model development using **TensorFlow and Keras**
- Data normalization and feature preparation
- Model evaluation using statistical metrics such as **MSE and R^2 Score**
- Development of interactive dashboards using **Streamlit**
- Data visualization using **Matplotlib**

Analytical and Soft Skills

- Problem-solving and analytical thinking
- Research and exploration of machine learning techniques
- Technical documentation and reporting
- Understanding real-world applications of Artificial Intelligence
- Project planning and execution

This project provided hands-on exposure to **building an end-to-end machine learning application**, from data collection to model deployment and visualization.

9. Testimonials from team

Working on this project provided valuable practical experience in applying machine learning and deep learning techniques to real-world financial data. It allowed us to understand how predictive models can be developed and integrated into interactive applications.

The project helped strengthen our knowledge in areas such as **data preprocessing, time-series analysis, model training, and dashboard development**. Through this experience, we improved our problem-solving abilities and gained confidence in implementing data-driven solutions.

Overall, this internship project was a meaningful learning experience that enhanced both our **technical knowledge and practical skills in Artificial Intelligence and Data Science**.

10. Conclusion

The project successfully demonstrates the application of **deep learning techniques for stock market prediction**. By using LSTM neural networks and historical stock price data, the system is capable of predicting future stock prices and visualizing trends through an interactive dashboard. The project integrates **data collection, preprocessing, model training, and visualization**, providing a complete AI-based solution for financial forecasting. Overall, this internship project enhanced practical knowledge of machine learning, data analysis, and software development.

11. Acknowledgements

I would like to thank **Infosys Springboard** for providing this valuable internship opportunity and learning platform. I also express my gratitude to the mentors and instructors for their guidance and support throughout the project. Their encouragement helped in successfully completing this internship and gaining practical knowledge in **Artificial Intelligence and Machine Learning**.